

Past Examination Question Repository

CS302: Advanced Data Structures & Algorithm Analysis — Full 2020–2025 Index

33

TOTAL QUESTIONS

500

TOTAL MARKS

6 Years

2020 – 2025

6

COURSE MODULES

MODULE 1: DYNAMIC PROGRAMMING (6 QUESTIONS)

Total: 110 Marks

YEAR	QUESTION #	TOPIC / SUB-TOPIC	QUESTION FOCUS & CORE PROBLEM STATEMENT	MARKS
2020	Q2(a)–(c)	Dynamic Programming	Maximum Subarray Sum (Kadane DP) & 2D Grid Optimal Path Cost	22 Marks
2021	Q1(b)	Dynamic Programming	Coin Change Problem: Minimum Coins & Total Unique Combinations	15 Marks
2022	Q3(a)	Dynamic Programming	Edit Distance (Levenshtein) with Variable Cost Substitution Matrix	20 Marks
2023	Q2(c)	Dynamic Programming	Longest Common Subsequence (LCS) Backtracking & String Reconstruction	15 Marks
2024	Q1(a)–(d)	Dynamic Programming	0/1 Knapsack Problem with Dual Volume & Weight Capacity Constraints	20 Marks
2025	Q3(a)–(b)	Dynamic Programming	Matrix Chain Multiplication: M-Table DP Recurrence & Parenthesization	18 Marks

MODULE 2: GRAPH ALGORITHMS & NETWORK FLOW (6 QUESTIONS)

Total: 106 Marks

YEAR	QUESTION #	TOPIC / SUB-TOPIC	QUESTION FOCUS & CORE PROBLEM STATEMENT	MARKS
2020	Q4(a)–(b)	Minimum Spanning Trees	Kruskal's vs Prim's Algorithms & Disjoint Set Union (Union-Find) Analysis	20 Marks
2021	Q3(a)–(c)	Graph Traversals	Breadth-First Search (BFS) & Depth-First Search (DFS) Edge Classifications	16 Marks
2022	Q2(b)	Shortest Paths	Floyd-Warshall All-Pairs Shortest Path Dynamic Distance Matrix Updates	18 Marks
2023	Q4(a)–(d)	Network Flow	Ford-Fulkerson Augmenting Paths, Residual Graphs & Min-Cut Extraction	25 Marks
2024	Q3(b)	Shortest Paths	Bellman-Ford Algorithm Execution & Negative Weight Cycle Proof	12 Marks
2025	Q1(c)–(d)	Shortest Paths	Dijkstra's Algorithm with Binary Heap vs Fibonacci Heap Time Bounds	15 Marks

MODULE 3: NP-COMPLETENESS & REDUCTIONS (6 QUESTIONS)

Total: 98 Marks

YEAR	QUESTION #	TOPIC / SUB-TOPIC	QUESTION FOCUS & CORE PROBLEM STATEMENT	MARKS
2020	Q5(c)	NP-Reduction Proof	Polynomial-Time Reduction from 3-SAT to Vertex Cover Problem	10 Marks
2021	Q4(a)–(b)	NP-Completeness	Clique to Independent Set Reduction & Polynomial Verification Proof	15 Marks
2022	Q5(a)–(c)	NP-Reduction Proof	Hamiltonian Cycle to Travelling Salesperson Problem (TSP) Reduction	20 Marks
2023	Q5(b)	NP-Completeness	Subset Sum Problem Reduction to 0/1 Knapsack NP-Completeness	15 Marks
2024	Q4(a)–(c)	NP-Reduction Proof	3-SAT Reduction to Graph 3-Colorability Problem Proof	20 Marks
2025	Q4(a)–(b)	NP-Completeness	Circuit-SAT to 3-SAT Transformation Steps & Equisatisfiability	18 Marks

MODULE 4: AMORTIZED ANALYSIS (6 QUESTIONS)

Total: 84 Marks

YEAR	QUESTION #	TOPIC / SUB-TOPIC	QUESTION FOCUS & CORE PROBLEM STATEMENT	MARKS
2020	Q1(b)	Amortized Methods	Aggregate Method vs Accounting Method for Dynamic Array Resizing	12 Marks
2021	Q2(a)	Potential Method	Potential Function Analysis for k-Bit Binary Counter Increments	15 Marks
2022	Q1(c)	Amortized Cost	Stack Operations with MULTIPOP Amortized Cost Analysis	12 Marks
2023	Q1(a)–(b)	Potential Method	Splay Tree Potential Function Analysis & Access Lemma Derivation	15 Marks
2024	Q2(c)	Potential Method	Dynamic Hash Table Expansion & Contraction Potential Analysis	15 Marks
2025	Q2(a)	Amortized Bounds	Fibonacci Heap Union & Extract-Min Amortized Complexity Analysis	15 Marks

Reduction Direction Invariant: In NP-Completeness questions (Module 3), marks are immediately lost if you reduce the new target problem into the known problem. Always reduce *Known NP-C Problem* $\leq_P$ *Target Problem*.

Potential Function Validity: When answering Module 4 questions, always prove that the potential function satisfies $\Phi(D_i) - \Phi(D_0) \geq 0$ for all $i \geq 1$ to establish an upper bound on actual costs.

MODULE 5: ADVANCED TREE STRUCTURES (5 QUESTIONS)

Total: 60 Marks

YEAR	QUESTION #	TOPIC / SUB-TOPIC	QUESTION FOCUS & CORE PROBLEM STATEMENT	MARKS
2020	Q3(a)	Red-Black Trees	Red-Black Tree Color Invariants, Black-Height Bounds & Rotations	10 Marks
2021	Q5(a)	B-Trees	B-Tree Node Insertion, Key Splitting Algorithm & Minimum Degree t Bounds	10 Marks
2022	Q4(a)–(b)	Splay Trees	Splay Tree Rotations: Zig-Zig, Zig-Zag, and Top-Down Splaying Traces	15 Marks
2023	Q3(b)	Balanced Trees	AVL Tree Height Balancing vs Red-Black Tree Strict Height Comparisons	10 Marks
2025	Q2(b)–(d)	B-Trees	B-Tree Key Deletion Cases: Underflow Borrowing & Node Merging Operations	15 Marks

MODULE 6: STRING MATCHING & APPROXIMATION (4 QUESTIONS)

Total: 42 Marks

YEAR	QUESTION #	TOPIC / SUB-TOPIC	QUESTION FOCUS & CORE PROBLEM STATEMENT	MARKS
2020	Q6(a)	String Matching	Rabin-Karp Rolling Hash Function & Spurious Hit Collision Resolution	10 Marks
2022	Q6(b)	String Matching	Knuth-Morris-Pratt (KMP) Prefix Automaton & π -Table Construction	10 Marks
2024	Q5(a)–(b)	Approximation Algorithms	2-Approximation Algorithm for Vertex Cover & Metric TSP Triangle Inequality	12 Marks
2025	Q5(c)	String Matching	Boyer-Moore Bad-Character Rule & Good-Suffix Shift Table Calculations	10 Marks

CHRONOLOGICAL EXAM PAPER SUMMARY (2020–2025)

Paper Totals

EXAM PAPER	INDEXED QUESTIONS	TOTAL PAPER MARKS	MODULES INCLUDED IN EXAM
2020 Paper	6 Sub-Questions	84 Marks	Modules 1, 2, 3, 4, 5, 6 (All Modules Covered)
2021 Paper	5 Sub-Questions	71 Marks	Modules 1, 2, 3, 4, 5
2022 Paper	6 Sub-Questions	95 Marks	Modules 1, 2, 3, 4, 5, 6 (All Modules Covered)
2023 Paper	5 Sub-Questions	80 Marks	Modules 1, 2, 3, 4, 5
2024 Paper	5 Sub-Questions	79 Marks	Modules 1, 2, 3, 4, 6
2025 Paper	6 Sub-Questions	91 Marks	Modules 1, 2, 3, 4, 5, 6 (All Modules Covered)

